

178. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8

178.1 Overview

This clause specifies the 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 PMDs and the associated baseband media. The PMDs provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links on one, two, four, or eight lanes, respectively, over an electrical backplane. The specifications for the four PMDs are similar, except for the number of lanes and associated parameters, and the MDI.

This clause defines specifications for two classes of transmitters and two classes of receivers, class A or class B, to allow for a range of package characteristics. A transmitter or receiver is classified as class A or class B if it meets all of the specifications for class A or class B, respectively. The required characteristics of the electrical interconnect between two PMDs depend on the transmitter class of one PMD and the receiver class of the other PMD.

The associated Annex 178A provides measurement methods for electrical channels and interfaces.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate sublayers (as specified in Table 178–1 through Table 178–4) and to the medium through the appropriate MDI, as illustrated in Figure 178–1.

Table 178–1—Physical Layer clauses associated with the 200GBASE-KR1 PMD

Associated clause	200GBASE-KR1
73—AN	Required
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120F—200GAUI-2 C2C ^c	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.5.1.

^c When implemented in a 200GBASE-KR1 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 200GBASE-KR1 PHY may include one instance of 200GAUI-n C2C as described in 176B.5.1.

Table 178–2—Physical Layer clauses associated with the 400GBASE-KR2 PMD

Associated clause	400GBASE-KR2
73—AN	Required
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.6.1.

^c When implemented in a 400GBASE-KR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 400GBASE-KR2 PHY may include one instance of 400GAUI-n C2C as described in 176B.6.1.

Table 178–3—Physical Layer clauses associated with the 800GBASE-KR4 PMD

Associated clause	800GBASE-KR4
73—AN	Required
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a

^a A 800GBASE-KR4 PHY may include one instance of 800GAUI-n C2C as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

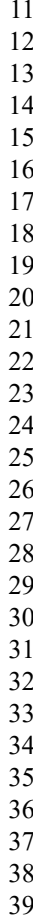
^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Table 178–4—Physical Layer clauses associated with the 1.6TBASE-KR8 PMD

Associated clause	1.6TBASE-KR8
73—AN	Required
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
170—1.6 Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a

^a A 1.6TBASE-KR8 PHY may include one instance of 1.6TAUI-n C2C as described in 176B.8.1.

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- 3
- 4

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With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 8×10^{-6} .

178.3 Conventions

This clause describes four PMDs, 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8, which have one, two, four, and eight lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 200GBASE-KR1, $n = 2$ for 400GBASE-KR2, $n = 4$ for 800GBASE-KR4, and $n = 8$ for 1.6TBASE-KR8.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 200GBASE-KR1 PMD, 400GBASE-KR2 PMD, 800GBASE-KR4 PMD, or 1.6TBASE-KR8 PMD.

Within this clause, the term FEC refers to the RS-FEC within the Clause 119 PCS for 200GBASE-KR1 and 400GBASE-KR2, the Clause 172 PCS for 800GBASE-KR4, and the Clause 175 PCS for 1.6TBASE-KR8.

178.4 Service interfaces

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (PMA immediately above the PMD). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-KR1 and 400GBASE-KR2, in 169.3 for 800GBASE-KR4, and in 174.3 for 1.6TBASE-KR8. The nominal signaling rate for each symbol stream in `PMD:IS_UNITDATA_i.request` and `PMD:IS_UNITDATA_i.indication`, where $i = 0$ to $n-1$, is 106.25 GBd.

The `SIGNAL_OK` parameter of the `PMD:IS_SIGNAL.indication` primitive (see 116.3.3.3) corresponds to the variable `rts_status` of the ready-to-send (RTS) function, as defined in 178B.8.2.1. When `SIGNAL_OK` is either `IN_PROGRESS` or `FAIL`, the `rx_symbol` parameters of `PMD:IS_UNITDATA_i.indication` on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the `PMD:IS_SIGNAL.request` primitive (see 116.3.3.4).

178.5 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive `AN_LINK.indication` defined in 73.9. The requirements are the same as those specified in 179.5.

178.6 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 178–5. It is assumed that the one-way delay through the medium is no more than 14 ns.

Descriptions of overall system delay constraints can be found in 116.4 for 200GBASE-KR1 and 400GBASE-KR2, in 169.4 for 800GBASE-KR4, and in 174.4 for 1.6TBASE-KR8.

Table 178–5—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GBASE-KR1	14 848	29	74.24
400GBASE-KR2	29 696	58	74.24
800GBASE-KR4	59 392	116	74.24
1.6TBASE-KR8	118 784	232	74.24

^a One bit time is equal to 5 ps for 200GBASE-KR1, 2.5 ps for 400GBASE-KR2, 1.25 ps for 800GBASE-KR4, and 625 fs for 1.6TBASE-KR8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GBASE-KR1, 1.28 ns for 400GBASE-KR2, 640 ps for 800GBASE-KR4, and 320 ps for 1.6TBASE-KR8. See 1.4.459 for the definition of pause_quanta.

178.7 Skew constraints

The Skew (relative delay) between the PCS lanes is kept within limits so that the information on the PCS lanes can be reassembled by the PCS. The Skew Variation is also limited to ensure that a given PCS lane always traverses the same physical lane.

178.7.1 Skew Constraints for 200GBASE-KR1 and 400GBASE-KR2

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5. Skew Variation at SP2 to SP5 is specified only for 400GBASE-KR2. For 200GBASE-KR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

178.7.2 Skew Constraints for 800GBASE-KR4 and 1.6TBASE-KR8

Skew and Skew Variation are defined in 169.5 for 800GBASE-KR4 and in 174.5 for 1.6TBASE-KR8. They are specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5 for 800GBASE-KR4, and in Figure 174–5 and Figure 174–6 for 1.6TBASE-KR8.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5 and 174.5.

178.8 PMD functional specifications

178.8.1 Reference test points

Reference test points are illustrated in Figure 178–2, which shows one direction of a 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, or 1.6TBASE-KR8 link. Additional test points for compliance measurement are defined in 178.9.

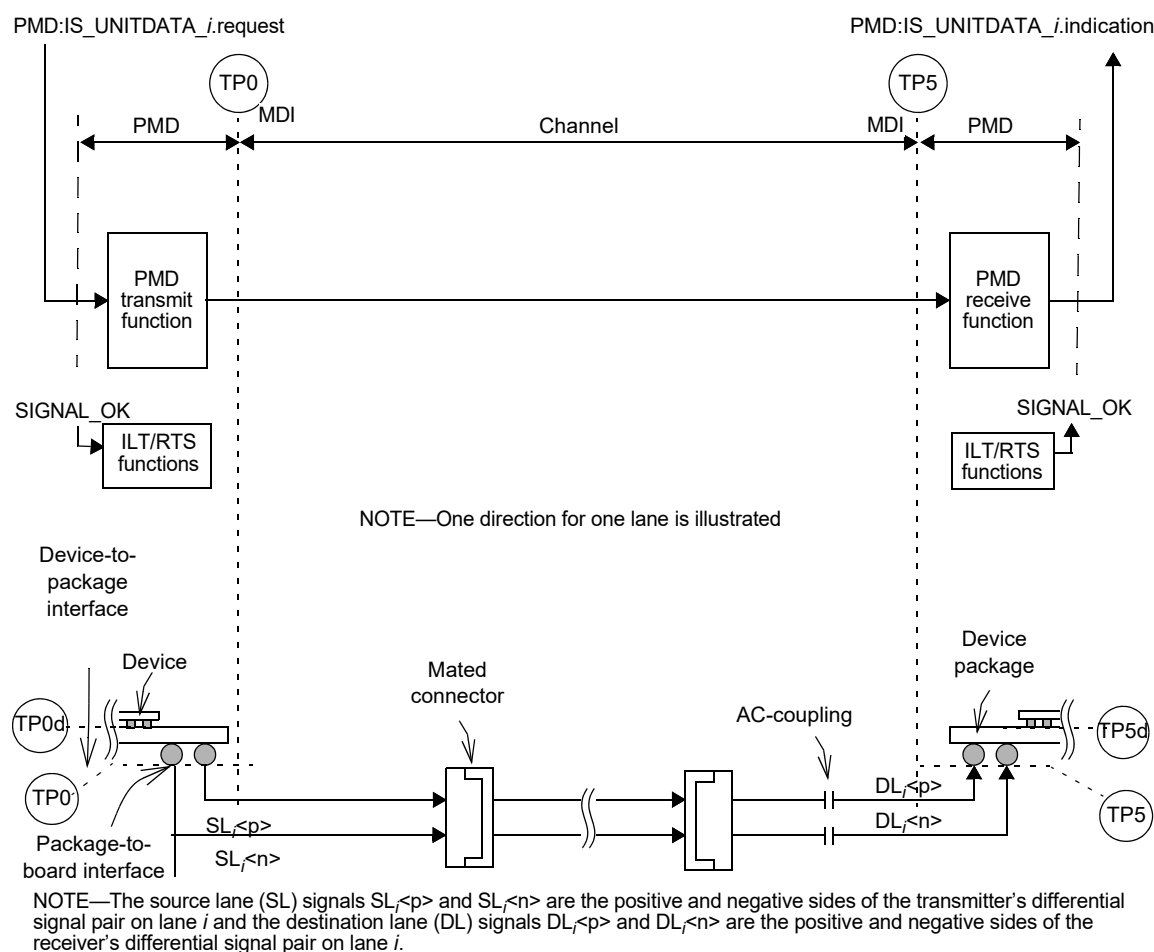


Figure 178–2—200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, or 1.6TBASE-KR8 link diagram

178.8.2 PMD transmit function

The PMD transmit function specification is identical to that of 179.8.2 with the exception that electrical signals are delivered to the MDI according to the transmit electrical specifications in 178.9.2.

178.8.3 PMD receive function

The PMD receive function specification is identical to that of 179.8.3 with the exception that electrical signals are received from the MDI according to the receive electrical specifications in 178.9.3.

178.8.4 PMD global signal detect function

The PMD global signal detect function specification is identical to that of 179.8.4.

178.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function specification is identical to that of 179.8.5.

178.8.6 PMD global transmit disable function

The PMD global transmit disable function is optional. Its specification is identical to that of 179.8.6.

178.8.7 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function is optional. Its specification is identical to that of 179.8.7.

178.8.8 PMD fault functions

The specification of PMD fault functions is identical to that of 179.8.8.

178.8.9 Autonomous path startup (APSU) functions

The PMD autonomous path startup (APSU) specification is identical to that of 179.8.9.

178.8.10 PMD reset function

The specification of the PMD reset function is identical to that of 179.8.10.

178.9 Electrical characteristics

178.9.1 Reference impedance

The reference impedance for differential specifications is $92.5\ \Omega$. The reference impedance for common-mode specifications is $23.125\ \Omega$.

NOTE—Calculation of modal S-parameters may require transformation of the measured single-ended S-parameters to the specified reference impedance. See 178A.1.3.

178.9.2 Transmitter characteristics

The transmitter on each lane shall meet the specifications at TP0v (see 178.9.2.1) given in Table 178–6 and detailed in the referenced subclauses.

Unless otherwise specified, transmitter or pattern generator signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP0v to 50 Ω single-ended loads in the test equipment.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP0v where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

Table 178–6—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate (range)	178.9.2.1	106.25 $\pm$ 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.2		
Transmitter disabled		0.03	V
Transmitter enabled		1	V
DC common-mode voltage (range)	176D.8.2	0.2 to 1	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	176D.8.2	0.032	V
Signal to AC common-mode noise ratio, $SCMR$ (min)	178.9.2.6	15	dB
Difference effective return loss, $dERL$ (min)	178.9.2.3	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	178.9.2.4	3.25	dB
Difference steady-state voltage, dv_f (range)	178.9.2.5	0 to 0.1	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	178.9.2.5	0	—
Common-mode to differential-mode return loss, RL_{dc} (min)	178.9.2.7	Equation (178–2)	
Level separation mismatch ratio, R_{LM} (min)	179.9.4.3	0.95	—
Transmitter output equalization			
absolute value of step size range for all taps	179.9.4.2.4	0.005 to 0.025	—
value at minimum state for $c(-3)$ (max)	179.9.4.2.5	–0.06	—
value at maximum state for $c(-2)$ (min)	179.9.4.2.5	0.12	—
value at minimum state for $c(-1)$ (max)	179.9.4.2.5	–0.34	—
value at minimum state for $c(0)$ (max)	179.9.4.2.5	0.5	—
value at minimum state for $c(1)$ (max)	179.9.4.2.5	–0.2	—
Signal-to-noise-and-distortion ratio, SNDR (min)	179.9.4.6	Table 179–9	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	179.9.4.4	28	dB
Output jitter (max)	179.9.4.7		
JH_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
JH_{4u}		0.118	UI

178.9.2.1 Transmitter signaling rate

The transmitter signaling rate shall be within the range specified in Table 178–6.

NOTE—If the Physical Layer includes an AUI-C2C interface, the signaling rate is derived from the signaling rate of that interface.

178.9.2.2 Transmitter test fixture

Unless otherwise noted, the transmitter is measured using the methodology described in Annex 163A at the output of a test fixture (TP0v) as shown in Figure 178–3. The transmitter test fixture is between TP0 and TP0v. Annex 163B gives example electrical characteristics of a test fixture for which reference values can be calculated.

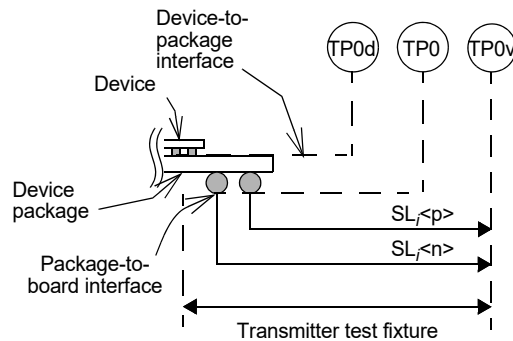


Figure 178–3—Transmitter test fixture and test points

178.9.2.2.1 Test fixture insertion loss

The insertion loss of the test fixture shall be between 3.4 dB and 8.5 dB at 53.125 GHz. It is recommended that the insertion loss of the test fixture be as low as feasible within the specified range. The magnitude of the insertion loss deviation of the test fixture shall be less than or equal to 0.2 dB from 0.05 GHz to 67 GHz. Insertion loss deviation is calculated as specified in 93A.4, where T_t is 0.005 ns, and f_b and f_r values are taken from Table 178–14.

178.9.2.2.2 Test fixture effective return loss (ERL)

The ERL of the test fixture at TP0v is computed using the procedure in 93A.5 with the values in Table 178–7 and with reference impedance equal to 92.5 Ω . Parameters that do not appear in Table 178–7 take values from Table 178–14.

Table 178–7—Test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Number of samples per unit interval	M	32	
Tukey window flag	tw	1	—

The ERL at TP0v shall be greater than or equal to 15 dB.

178.9.2.2.3 Test fixture common-mode to common-mode return loss

The common-mode to common-mode return loss of the test fixture shall be greater than or equal to 6 dB at all frequencies between 0.2 GHz and 67 GHz.

178.9.2.3 Transmitter difference ERL

The difference ERL, $dERL$, of the transmitter at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 178–8, Table 178–14, and Table 178–15 with differential reference impedance equal to $92.5\ \Omega$, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v. The reference value, $ERL^{(ref)}$, is calculated using the package model of the transmitter class to which the PMD adheres.

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) in Table 178–6.

Table 178–8—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Number of samples per unit interval	M	32	
Tukey window flag	tw	1	—

178.9.2.4 Transmitter common-mode to common-mode return loss

The common-mode to common-mode return loss shall be greater than or equal to RL_{cc} (min) in Table 178–6 at all frequencies between 0.2 GHz and 67 GHz.

178.9.2.5 Difference steady-state voltage and difference linear fit pulse peak ratio

The difference steady-state voltage dv_f , and difference linear fit pulse peak ratio dR_{peak} are computed using the procedure in 163A.3.2.1 with $N_v = 400$, $M = 32$, $D_p = 4$, and other parameter values specified in Table 178–14 and Table 178–15 with the following exceptions:

- The linear fit pulse is computed using the procedure defined in 179.9.4.2.1.
- The value of R_0 is $50\ \Omega$.
- The measured TP0 to TP0v channel S-parameters are normalized to $50\ \Omega$.
- The value of f_r is 60 GHz.

The reference steady-state voltage $v_f^{(ref)}$ and reference value $R_{peak}^{(ref)}$ are calculated using the device package model of the transmitter class to which the PMD adheres.

The difference steady-state voltage at TP0v shall be within the range specified for dv_f in Table 178–6.

The difference linear fit pulse peak ratio at TP0v shall meet the specification dR_{peak} (min) in Table 178–6.

178.9.2.6 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is calculated using Equation (178–1).

$$SCMR = 10\log_{10}\left(\frac{P_{Signal}}{VCM_{FB}^2}\right) \quad (178-1)$$

where

$SCMR$	is the signal to AC common-mode noise ratio in dB
P_{Signal}	is defined in 179.9.4.6
VCM_{FB}	is the full-band peak-to-peak AC common-mode voltage defined by the method specified in 176D.8.2 and measured with the transmitter equalization set to NO EQUALIZATION

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 178–6.

178.9.2.7 Transmitter common-mode to differential-mode return loss

The common-mode to differential-mode return loss of the transmitter shall meet Equation (178–2). The common-mode to differential-mode return loss limit is illustrated by Figure 178–4.

$$RLdc(f) \geq \begin{cases} 25 - 20\frac{f}{106.25} & 0.05 \leq f \leq 53.125 \\ 15 & 53.125 \leq f \leq 67 \end{cases} \quad (178-2)$$

where

$RLdc$	is the common-mode to differential-mode return loss in dB
f	is the frequency in GHz

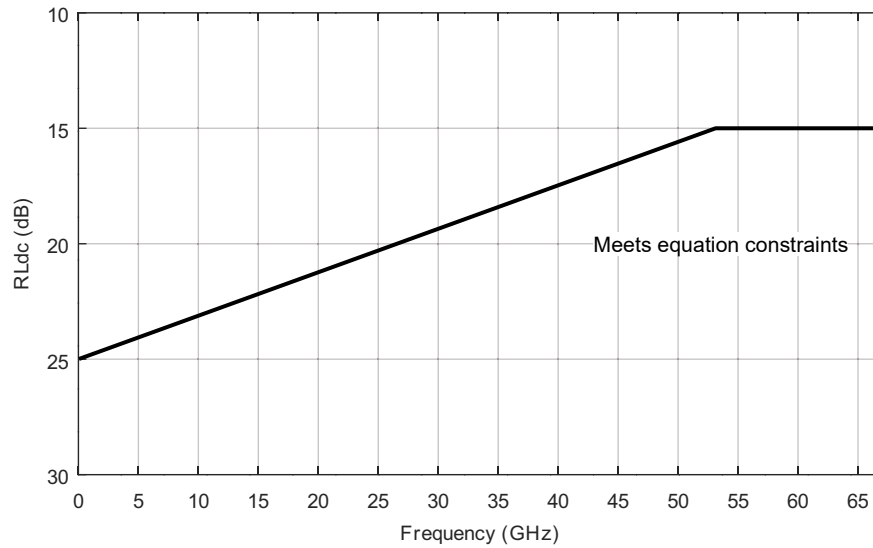


Figure 178-4—Transmitter common-mode to differential-mode return loss

178.9.3 Receiver characteristics

The receiver on each lane shall meet the specifications at TP5v given in Table 178-9 and detailed in the referenced subclauses.

Table 178-9—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate (range)	178.9.3.2	106.25 ± 50 ppm	GBd
Transmitter output amplitude tolerance	178.9.3.6	0.5	V
Interference tolerance	178.9.3.4	Table 178-12	—
Jitter tolerance	178.9.3.5	Table 179-16	—
Difference effective return loss, $dERL$ (min)	178.9.3.7	-3	dB
Differential-mode to common-mode return loss, RL_{cd}	178.9.3.8	Equation (178-5)	dB

178.9.3.1 Receiver test fixture

Unless otherwise noted, measurements of the receiver are made at the input of a test fixture (TP5v) as shown in Figure 178–5. The receiver test fixture is between TP5 and TP5v.

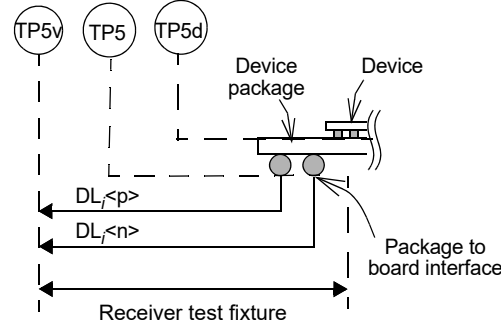


Figure 178–5—Receiver test fixture and test points

Receiver test fixture characteristics are the same as specified in 178.9.2.2.

178.9.3.2 Receiver signaling rate

A PHY shall comply with the receiver requirements of 178.9.3.4 through 178.9.3.6 for any signaling rate in the range specified in Table 178–9.

178.9.3.3 Error ratio requirements for receiver tests

A receiver on each lane is required to meet the expected BLER specified in 178.2, using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 178–10. The error mask $H_{\max}(k)$ to be used in the method of 174A.9.5 is provided in Table 178–11.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m^{(i)}(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m^{(i)}(k))$ for $k = n-2$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $[\log_{10}(H_m^{(i)}(k)) - \log_{10}(H_m^{(i)}(4))] / (k - 4)$ should be less than $[\log_{10}(H_m^{(i)}(4)) - \log_{10}(H_m^{(i)}(2))] / 2$ for each k greater than 4.

Table 178–10—Error Ratio Parameters

Parameter name	Value
p	
200GBASE-KR1	1
400GBASE-KR2	2
800GBASE-KR4	4
1.6TBASE-KR8	8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	1.6×10^{-5}
BER_{max}	2.761×10^{-4}

Table 178–11—Receiver Error Mask

Test symbol errors per test block, k (see 174A.9.5)	Probability $H_{\max}(k)$			
	$p = 1$	$p = 2$	$p = 4$	$p = 8$
1	3.3×10^{-1}	3.5×10^{-1}	2.6×10^{-1}	1.6×10^{-1}
2	2.5×10^{-1}	1.3×10^{-1}	4.8×10^{-2}	1.4×10^{-2}
3	1.3×10^{-1}	3.3×10^{-2}	6×10^{-3}	8.8×10^{-4}
4	4.7×10^{-2}	6.2×10^{-3}	5.5×10^{-4}	3.9×10^{-5}
5	1.4×10^{-2}	9.1×10^{-4}	4×10^{-5}	1.4×10^{-6}
6	3.5×10^{-3}	1.1×10^{-4}	2.4×10^{-6}	4.1×10^{-8}
7	7.4×10^{-4}	1.2×10^{-5}	1.2×10^{-7}	9.9×10^{-10}
8	1.4×10^{-4}	1.1×10^{-6}	5.5×10^{-9}	2.1×10^{-11}
9	2.3×10^{-5}	8.8×10^{-8}	2.2×10^{-10}	3.9×10^{-13}
10	3.4×10^{-6}	6.4×10^{-9}	7.6×10^{-12}	6.3×10^{-15}
11	4.5×10^{-7}	4.2×10^{-10}	2.4×10^{-13}	9.2×10^{-17}
12	5.5×10^{-8}	2.5×10^{-11}	7×10^{-15}	1.2×10^{-18}
13	6.3×10^{-9}	1.4×10^{-12}	1.8×10^{-16}	1.4×10^{-20}
14	6.6×10^{-10}	7.2×10^{-14}	4.5×10^{-18}	1.6×10^{-22}
15	6.4×10^{-11}	3.4×10^{-15}	1×10^{-19}	1.6×10^{-24}
16	5.9×10^{-12}	1.5×10^{-16}	2.1×10^{-21}	1.4×10^{-26}

178.9.3.4 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 178.9.3.4.1 through 178.9.3.4.3.

Receiver interference tolerance is specified by two tests defined in Table 178–12. Test L includes a low-loss channel. Test H includes a high-loss channel.

NOTE—The channel for Test H is dependent on the receiver class that a PMD under test complies with.

A receiver shall meet the requirements of this test under both Test L and Test H conditions.

Table 178–12—Receiver interference tolerance parameters

Parameter	Test L (low loss)		Test H (high loss)			Units
	Max	Target	Min	Max	Target	
BLER ^a	$< 1.45 \times 10^{-11}$					—
Insertion loss, IL_{dd} , at 53.125 GHz ^b		N/A ^c				
Receiver class A	18		33.5	34.5	—	dB
Receiver class B	18		30	31	—	dB
COM including effects of broadband noise ^d		3	—	—	3	dB

^aThe BLER (see 178.2) is measured instead of the FEC symbol error ratio in step 10) of the receiver interference tolerance method defined in 93C.2.

^b IL_{dd} measured between TPt and TP5 (see Figure 93C–4) plus IL_{dd} of the specific package used by the test transmitter. See 178.9.3.4.2.

^cFor test L, the connection between TP0v and TP5v (see Figure 93C–2) is a noise coupler without an additional ISI channel, and the insertion loss is not specified.

^dThe COM value is the target for the receiver noise level calibration defined in 178.9.3.4.2. The channel noise voltage applied in 178.9.3.4.2 should be as close as practical to the value needed to produce the target COM. If higher noise values are used, this would demonstrate margin to the specification but this is not required for compliance.

178.9.3.4.1 Test setup

The interference tolerance test is performed with the setup shown in Figure 93C–2 or its equivalent. Characterization and calibration of the various elements in the test setup uses the test configurations in Figure 93C–3, Figure 93C–4, Figure 93C–5, and Figure 93C–6. In each figure TP0v replaces TP0a and TP5v replaces TP5a. For Test L in Table 178–12, the ISI channel in Figure 93C–2 and Figure 93C–4 is omitted.

The transmitter meets the requirements stated in 178.9.2 with the added constraint that the difference steady-state voltage dV_F (see 178.9.2.5) is less than or equal to the minimum specified in Table 178–6. The ISI channel emulates the frequency-dependent loss of the channel. The channel noise source represents non-equalizable impairments that may be introduced by a transmitter or channel.

The ERL of the test setup in Figure 93C–4 measured at TP5 replica towards TPt meets the requirements in 178.10.3.

The channel noise source has an adjustable output amplitude that is set according to the calibration procedure in 178.9.3.4.2. The noise produced by the channel noise source is measured directly at the output of the noise source (see Figure 93C–6). The noise is Gaussian with a crest factor of at least 5. The noise spectral density, $NSD(f)$, is normalized and constrained according to the relations in Equation (93C-1), where f_b is the signaling rate and f_{NSDI} is 1 GHz. The average noise spectral density, $NSD_{average}$, is defined by Equation (93C-2).

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used to measure the output of the transmitter and the channel noise source.

178.9.3.4.2 Noise calibration

The output amplitude of channel noise source is set to yield a target test channel Channel Operating Margin (COM). The test channel COM is derived from the S-parameters for the test channel from TPt to TP5 replica

$S^{(c)}$ measured using the test setup in Figure 93C-4. COM is calculated according to Annex 178A using the parameter values in Table 178-14 and Table 178-15 for the corresponding receiver class but with the exceptions contained in the remainder of this subclause.

The transmitter model is determined in one of the following ways.

- If the transmitter is a device with known S-parameters and transition time T_r , these parameters are used instead of the transmitter device package model in 178A.1.4, and the $ILdd$ in Table 178-12 is the $ILdd$ of the concatenated S-parameters of the device and the channel between TPt and TP5 replica. T_r should be provided as the value at the input of the device S-parameters network, as defined in 120G.3.1.4 but with no observation filter.
- If a calibrated instrument-grade transmitter is used, the TP0 to TP0a trace and the ISI channel in Figure 93C-2 and Figure 93C-4 are omitted, the TP0 to TP0a replica trace in Figure 93C-3 is omitted, and the transmitter model $S^{(t)}$ is omitted from Equation (178A-2). In this case, the $ILdd$ in Table 178-12 is the $ILdd$ of the channel between TPt and TP5 replica. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A-13) uses the filter $H_i(f)$ defined in 178A.1.6.2, where T_r is the measured transmitter transition time (see 120G.3.1.4).

In addition, the following parameters are derived from measurements of the test transmitter collected using the test setup in Figure 93C-3.

- The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 85$.
- The parameter R_{LM} is set to the measured value of R_{LM} .
- The parameters A_{DD} and σ_{RJ} are calculated from the measured values of JH4u and JH_{RMS} using Equation (178-3) and Equation (178-4) respectively. If the transmitter jitter can be controlled, it is recommended to adjust jitter such that JH4u and JH_{RMS} are as close as practical to their limits in Table 178-6.

The interference noise (see 178A.1.5) is derived from the S-parameters for the noise path from TPn to TP5 replica $S^{(n)}$ measured using the test setup shown in Figure 93C-5. For a given RMS noise amplitude σ_{ns} , the spectral density of the channel noise source $N_s(f)$ may be assumed to be $2\sigma_{ns}^2/f_b$ from DC to $f_b/2$ and 0 for frequencies greater than $f_b/2$. The spectral density measured at the output of the channel noise source, $NSD(f)$, may also be used as $N_s(f)$ for the calculation of COM and, in such cases, the value of σ_{ns}^2 is the integral of $NSD(f)$ over frequency.

COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 178-14 on the receiver side. The value of COM is taken as the lower of the two calculated values. The value of σ_{ns} that yields the target value of COM specified in Table 178-12 is determined (via iterative calculation or by other means). The amplitude of the channel noise source is then configured so that the measured RMS value at its output matches σ_{ns} .

$$A_{DD} = \frac{\frac{JH4u}{2} + Q4d \sqrt{(Q4d^2 + 1) \times JH_{RMS}^2 - \left(\frac{JH4u}{2}\right)^2}}{Q4d^2 + 1} \quad (178-3)$$

$$\sigma_{RJ} = \frac{\frac{JH4u}{2} - A_{DD}}{Q4d} \quad (178-4)$$

where

$$Q4d = 3.719$$

NOTE 1— $Q4d$ is an approximated solution of $Q(Q4d) = 1 \times 10^{-4}$, where the Q function is defined in Equation (95-1).

NOTE 2—Calculation of A_{DD} requires that $(Q4d^2 + 1) \times JH_{RMS}^2 \geq \left(\frac{JH4u}{2}\right)^2$. If this does not hold, a different transmitter should be used in the test setup.

178.9.3.4.3 Test procedure

The receiver of each lane is tested using the following procedure.

- a) Calibrate the amplitude of the channel noise source using the procedure defined in 178.9.3.4.2.
- b) The receiver under test configures the test transmitter equalizer coefficients and precoding to the settings it would select using the ILT function described in 178.8.9.
- c) Configure the test transmitter to transmit either scrambled idle or PRBS31Q. During the test, the transmitters in all lanes of the PMD under test transmit scrambled idle or PRBS31Q, with equalization turned off (PRESET 1 condition).

A receiver shall meet the error ratio requirements defined in 178.9.3.3 under the conditions specified for this test for the receiver class that the PMD complies with, for both Test L and Test H in Table 178–12.

178.9.3.5 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 178.9.3.5.1 and 178.9.3.5.2.

178.9.3.5.1 Test setup

The test setup shown in Figure 178–6, or its equivalent, is used. The test channel meets the insertion loss requirement for Test H in Table 178–12.

178.9.3.5.2 Test procedure

The jitter tolerance test procedure is similar to that of 178.9.3.4, with the following exceptions:

- For each case in Table 179–16, the synthesizer is set to the specified frequency and the synthesizer output amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency measured at TP0v.
- The test channel calibration (see 178.9.3.4.2) is performed with the transmitter output set to the jitter frequency and amplitude for Case G of Table 179–16. The additive noise level obtained is used for all other cases.

A receiver shall meet the error ratio requirement defined in 178.9.3.3 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179–16.

178.9.3.6 Transmitter output amplitude tolerance

Transmitter output amplitude tolerance of a receiver is defined as the maximum transmitter steady-state voltage that the receiver can tolerate, using the Test L (low loss) test channel specified in Table 178–12, such that it satisfies the error ratio requirements defined in 178.9.3.3 when the PMD transmit function operates in DATA mode (see 179.8.2).

The transmitter steady-state voltage is measured as specified in 179.9.4.2.2 at the output of the pattern generator used in the test. The differential peak-to-peak output voltage of the pattern generator does not exceed the maximum value specified in Table 178–6.

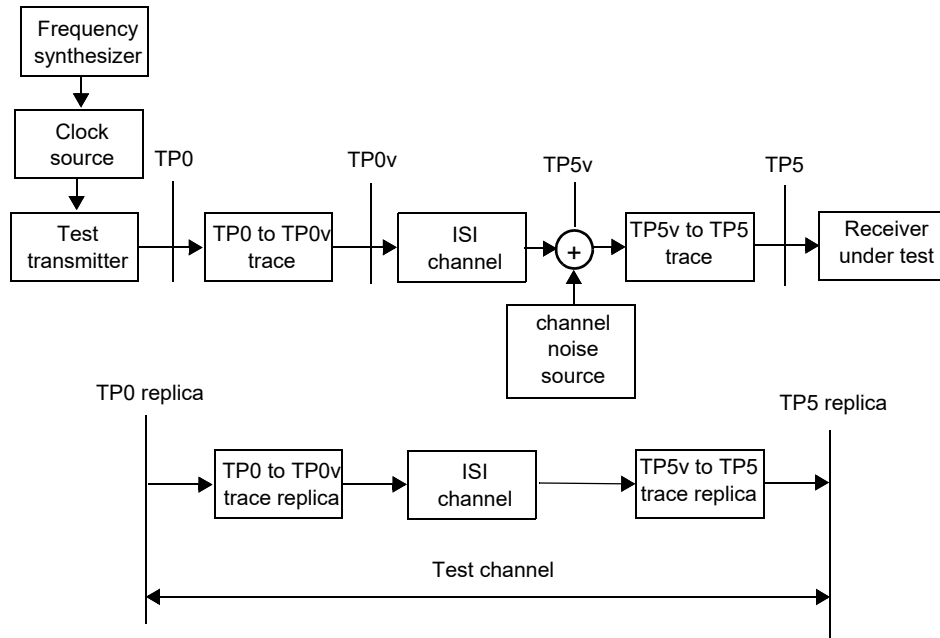


Figure 178-6—Jitter tolerance test setup

The pattern generator is initially configured to transmit training frames as defined in 178B.7.3 with transmit equalization at the INITIALIZE setting (see Table 179-8). During the initialization period, the device under test (DUT) may further configure the pattern generator to the transmit equalizer and precoding settings it would select using the ILT function (see 178.8.9). The settings may be communicated via the ILT function or by other means. After the pattern generator has been configured, the BLER is measured as specified in 174A.9 or 174A.11.

NOTE—Steady-state voltage is defined with PRESET 1. It is not initially generated by a transmitter, due to the INITIALIZE setting in Table 179-8. The receiver is not required to tolerate PRESET 1 unless it specifically requests it. The voltage observed at the receiver input is further attenuated by the test channel. Therefore, the receiver is not required to tolerate the maximum transmitter output voltage at its input.

178.9.3.7 Receiver difference ERL

The difference ERL of the receiver at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 178-8, Table 178-14, and 178-15 and with the value of T_{fx} equal to twice the delay from TP5v to TP5. The reference value, $ERL^{(ref)}$, is calculated using the device package model of the receiver class to which the PMD adheres.

The receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 178-9.

178.9.3.8 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (178-5). The differential-mode to common-mode return loss limit is illustrated by Figure 178-7.

$$RL_{cd}(f) \geq \begin{cases} 25 - 20 \frac{f}{106.25} & 0.05 \leq f \leq 53.125 \\ 15 & 53.125 \leq f \leq 67 \end{cases} \quad (178-5)$$

where

$RL_{cd}(f)$ is the differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

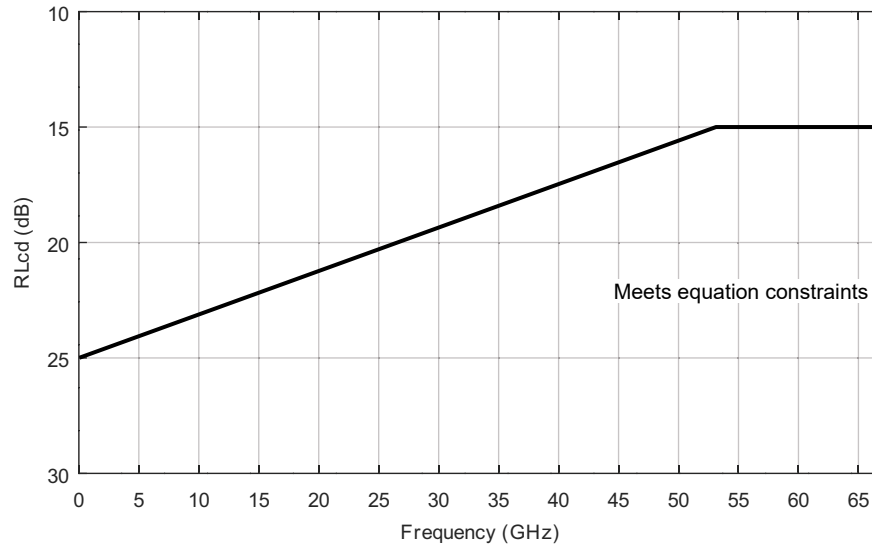


Figure 178-7—Receiver differential-mode to common-mode return loss limit

178.10 Channel characteristics

Table 178-13 provides a summary of the channel characteristics for 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 and references to the subclauses addressing each parameter. Unless otherwise indicated, the channel is defined between TP0 and TP5.

Table 178-13—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	178.10.1	3	dB
Maximum insertion loss from TP0d to TP5d, IL_{dd} , at 53.125 GHz (recommended)	178.10.2	40	dB
Minimum channel ERL ^a	178.10.3.1	11	dB
Differential-mode to common-mode return loss	178.10.3.3	Table 178-17	dB
Maximum AC-coupling 3 dB corner frequency	178.10.4	250	kHz
Minimum channel signal to common-mode ratio, $SCMR_{CH}$	178.10.5	17	dB

^aChannels with COM greater than 4 dB are not required to meet minimum ERL.

178.10.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 178A.1 and the values in Table 178–14 and Table 178–15. COM is calculated with the parameters for the transmitter and receiver classes that the channel under test is intended to support. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line stage 1 length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2.

NOTE—The parameters and values in Table 178–14 and Table 178–15 correspond to behavioral models of transmitters and receivers that are compliant to the PMD specifications in this clause. The purpose of these parameters and values is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementation. It is expected that the PMD specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation.

Table 178–14—Device termination and package model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1	$z_p^{(1)}$	33	mm
Transmission line 1 length, Test 2	$z_p^{(1)}$	12	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1, Tx / Rx	$z_p^{(1)}$	45 / 44	mm
Transmission line 1 length, Test 2, Tx / Rx	$z_p^{(1)}$	30 / 29	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Single-ended reference resistance	R_0	46.25	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package model parameters are based on the reference impedance R_0 .			

Table 178–15—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—
Transmitter equalizer, coefficient –2	$c(-2)$	0	—
Minimum value		0.14	
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient –1	$c(-1)$	–0.34	—
Minimum value		0	
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient 0	$c(0)$	0.54	—
Minimum value			
Transmitter equalizer, coefficient 1	$c(1)$	–0.2	—
Minimum value		0	
Maximum value		0.02	
Step size			
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	7.5×10^{-9}	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Number of samples per unit interval	M	32	
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI

Table 178–15—COM parameter values

Parameter	Symbol	Value	Units
Noise-equivalent quantization bits	N_{qb}	6	
Quantization clip probability	P_{qc}	10^{-7}	
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	6	—
Number of fixed-position taps ^a	N_{fix}	15	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index ^b	N_{max}	87	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	−0.7 −0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-4}	—
Include maximum likelihood sequence detection		yes	—

^a The number of fixed-position post-cursor taps is 8.

^b The latest post-cursor position for a floating tap is 80.

178.10.2 Channel insertion loss (recommended)

The recommended maximum insertion loss, IL_{dd} , for the channel between TP0d and TP5d is 40 dB at 53.125 GHz.

178.10.3 Channel return loss specifications

178.10.3.1 Channel ERL

The ERL of the channel at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 178–15 and Table 178–16, and with differential reference impedance equal to 92.5 Ω .

Channel ERL at TP0 and at TP5 shall be greater than or equal to 11 dB.

Table 178–16—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	7000	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

178.10.3.2 Channel modal ERL

Channel modal ERL (ERL_{CD}) is the ERL computed according to 93A.5 with the following exceptions:

- For channel input ERL_{CD} , $S_{ii}(f)$ (see 93A.5.1) is replaced with the channel $S_{CD11}(f)$.
- For channel output ERL_{CD} , $S_{ii}(f)$ is replaced with the channel $S_{CD22}(f)$.

The values of $S_{CD11}(f)$ and $S_{CD22}(f)$ are defined in 178A.1.3. Modal ERL is computed using the values in Table 178–15 and Table 178–16. The differential reference impedance is 92.5 Ω and the common-mode reference impedance is 23.125 Ω .

178.10.3.3 Channel modal return loss

The channel modal return loss specification is differential-mode to common-mode.

The TP0 to TP5 channel differential-mode to common-mode return loss is satisfied by either meeting the requirement of Equation (178–6) or by having ERL_{CD} higher than the value in Table 178–17.

$$RL_{cd}(f) \geq \begin{cases} 22 - 10 \frac{f}{53.125} & 0.05 \leq f < 53.125 \\ 15 - 3 \frac{f}{53.125} & 53.125 \leq f \leq 60 \end{cases} \quad (178-6)$$

where

- $RL_{cd}(f)$ is the negative of the value in dB of $s_{CD11}(f)$ defined in 176D.8.3.3
- f is the frequency in GHz

The TP0 to TP5 channel differential-mode to common-mode return loss limit is illustrated in Figure 178–8.

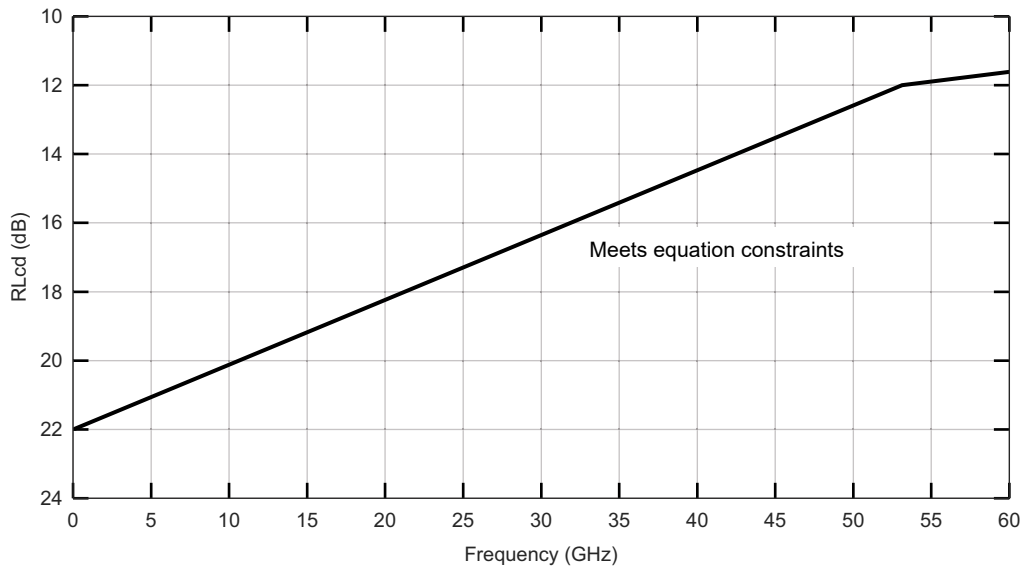


Figure 178–8— Channel differential-mode to common-mode return loss mask

Table 178–17—Channel modal return loss specifications

RL mask equation	RL mask figure	ERL_{CD} minimum (dB)
Equation (178–6)	Figure 178–8	17

178.10.4 AC-coupling

AC-coupling shall be implemented within the channel using DC blocking capacitors. The low-frequency 3 dB cutoff of the channel shall be less than 250 kHz.

Systems with no AC-coupling within the channel are considered engineered links. It is the system integrator’s responsibility to verify that the transmitter and the receiver are compatible with the common-mode voltage differences that may exist in this configuration.

178.10.5 Channel signal to common-mode ratio

Channel signal to common-mode ratio, $SCMR_{CH}$, is calculated using the procedure in 178A.2 with $P_{peak} = 10^{-7}$ and other parameter values taken from Table 178–15. $SCMR_{CH}$ shall be greater than or equal to 17 dB.

178.11 MDI specifications

The MDI for the 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 178–2.

178.12 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 93.10.

178.13 Management variables

The PMD control and status variables are identical to those defined in 179.14 with the exception that references to CR PMDs are replaced by references to the equivalent KR PMDs.

178.14 Protocol implementation conformance statement (PICS) proforma for Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8⁷

178.14.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

178.14.2 Identification

178.14.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

178.14.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

178.14.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*PMD	PMD or PHY	178.1	Item marked with PMD are PMD or PHY specifications, these items are not applicable to cable assemblies.	O.1	Yes [] No []
*CHNL	Channel	178.10	Item marked with CHNL are channel specifications, these items are not applicable to PMDs or PHYs.	O.1	Yes [] No []
KR1	200GBASE-KR1 PMD	178.1	Can operate as a 200GBASE-KR1 PMD	PMD:O.2	Yes [] No []
KR2	400GBASE-KR2 PMD	178.1	Can operate as a 400GBASE-KR2 PMD	PMD:O.2	Yes [] No []
KR4	800GBASE-KR4 PMD	178.1	Can operate as a 800GBASE-KR4 PMD	PMD:O.2	Yes [] No []
KR8	1.6TBASE-KR8 PMD	178.1	Can operate as a 1.6TBASE-KR8 PMD	PMD:O.2	Yes [] No []
PHY	Single package PHY	178.9.2	PMD and PCS are in the same package	PMD:O/3	Yes [] No []
AUIC2C	200 Gb/s per lane AUI-n C2C	178.1	PMD is separated from the PCS by an AUI-C2C specified in Annex 120B, Annex 120D, Annex 120F, or Annex 176C	PMD:O/3	Yes [] No [] N/A []
nGMII	200GMII, 400GMII, 800GMII, or 1.6TMII	178.1	Interface is implemented or functionally equivalent	PMD:O	Yes [] No [] N/A []
AN	Auto-Negotiation	178.1	PMD implements Auto-Negotiation	PMD:M	Yes []
PMDAPSU	APSU functions in PMD	178.8.9	APSU functions are implemented in the PMD	PMD:M	Yes [] N/A []
*MD	MDIO capability	178.13	PMD implements Clause 45 MDIO	PMD:O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

178.14.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8

178.14.4.1 PHY Specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PS1	Service Interface	178.4	PMD service interface implemented	PMD:M	Yes [] N/A []
PS2	Delay constraints	178.6	Complies with the delay constraints	PMD:M	Yes [] N/A []
PS3	Skew constraints	178.7	Complies with the skew constraints	PMD:M	Yes [] N/A []
PS4	PMD functional specifications	178.8	Complies with all PMD functional specifications	PMD:M	Yes [] N/A []
PS5	PMD electrical specifications	178.9	Complies with all PMD electrical specifications	PMD:M	Yes [] N/A []

178.14.4.2 Channel characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
CH1	Channel type 200GBASE-KR1	178.10	Supports single-lane links between two 200GBASE-KR1 PHYs	CHNL:O /4	Yes [] No [] N/A []
CH2	Channel type 400GBASE-KR2	178.10	Supports two-lane links between two 400GBASE-KR2 PHYs	CHNL:O /4	Yes [] No [] N/A []
CH3	Channel type 800GBASE-KR4	178.10	Supports four-lane links between two 800GBASE-KR4 PHYs	CHNL:O /4	Yes [] No [] N/A []
CH4	Channel type 1.6TBASE-KR8	178.10	Supports eight-lane links between two 1.6TBASE-KR8 PHYs	CHNL:O /4	Yes [] No [] N/A []
CH5	Channel electrical specifications	178.10	Complies with all channel electrical specifications	CHNL:M	Yes [] No [] N/A []

178.14.4.3 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	178.12	Conform to applicable requirements of 93.10	M	Yes []

178.14.4.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	178.13		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	178.13		MD:M	Yes [] N/A []

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